ABSTRACT

The increasing availability of aircraft surveillance data has created opportunities for developing systems that can monitor, analyze, and visualize air traffic in a more accessible manner. Automatic Dependent Surveillance–Broadcast (ADS-B) is a modern surveillance technology in which aircraft periodically broadcast information such as identification, position, altitude, velocity, and heading. These broadcasts can be received by ground stations and processed to provide real-time awareness of aircraft movements. ADS-B has become an important component of contemporary air traffic monitoring due to its ability to provide accurate and continuously updated flight information.

This project focuses on the design and development of an ADS-B Live Aircraft Tracking System using PingStation data as the primary source of surveillance information. The objective of the system is to transform raw ADS-B messages into meaningful aircraft information that can be monitored, stored, and analyzed through an intuitive web-based interface. Instead of relying on specialized aviation software, the project provides a lightweight and self-contained platform capable of handling aircraft tracking operations using commonly available technologies.

The system receives either live or previously captured ADS-B data streams and processes the incoming messages through a custom decoder developed for PingStation binary frames. The decoder extracts relevant aircraft parameters such as ICAO aircraft address, callsign, latitude, longitude, altitude, speed, heading, and timestamp information. After decoding, the data is validated and normalized to ensure consistency across different message types and transmission intervals. This processed information is then stored in a structured database for efficient retrieval and historical analysis.

The backend of the application is implemented using Python and Flask, providing APIs and services responsible for data ingestion, processing, storage, and communication with the frontend. SQLite is used as the primary local database for maintaining aircraft records and trajectory history, while CSV and Excel export functionalities are incorporated to support external analysis and reporting. These